

① Three main pillars in Machine learning

1.



Linear Algebra

Structure / Relationship between data



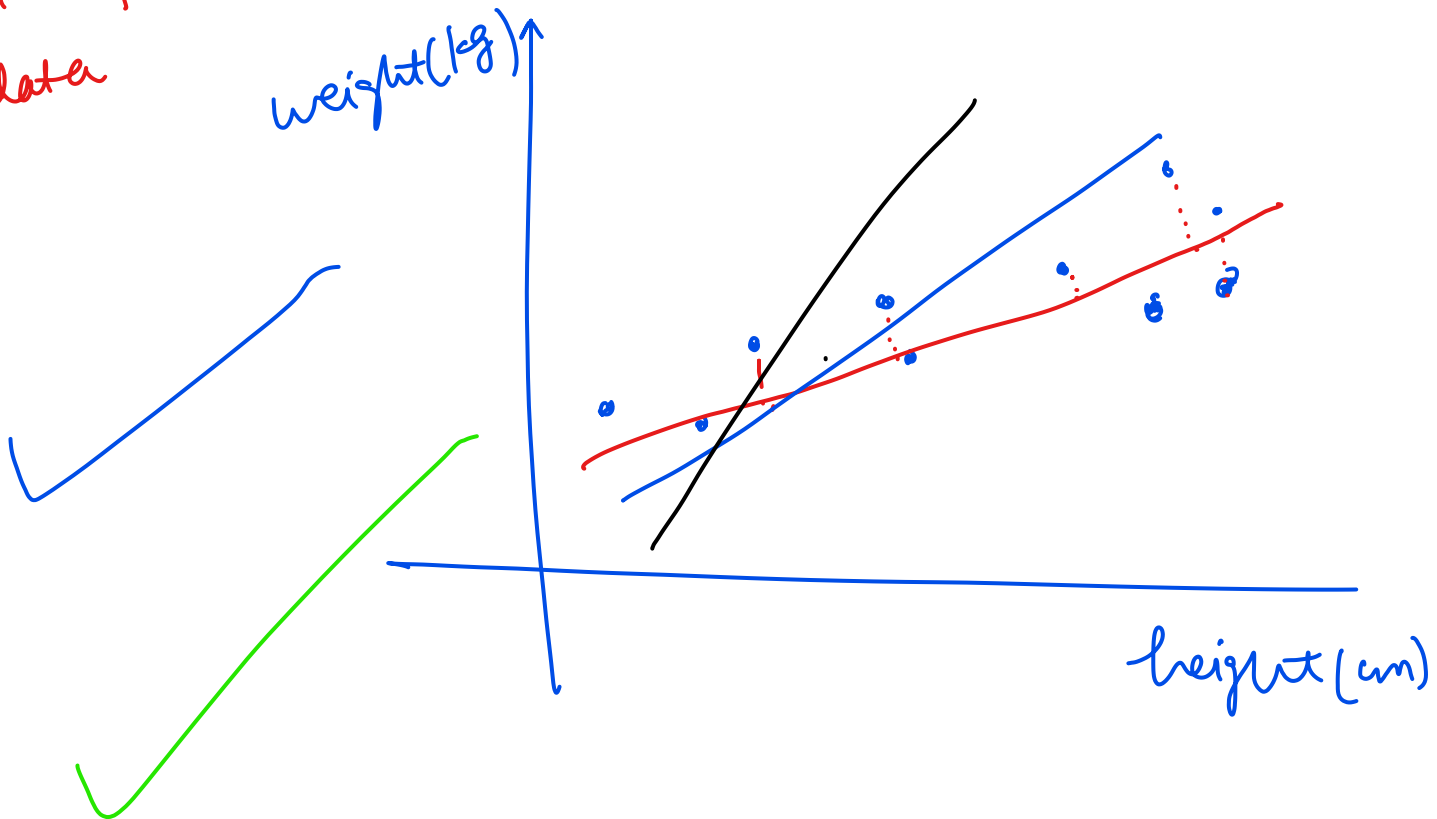
Probability

Noise / Uncertainty



Optimization

Data → decisions



Optimization:

→ Google maps finds the shortest route.

→ Uber sets optimal prices to balance supply & demand.

is about finding the best solution of a function.

→ A delivery company wants the quickest route to deliver packages (minimizing travel time).

→ "best" classifier.

↳ "least" loss

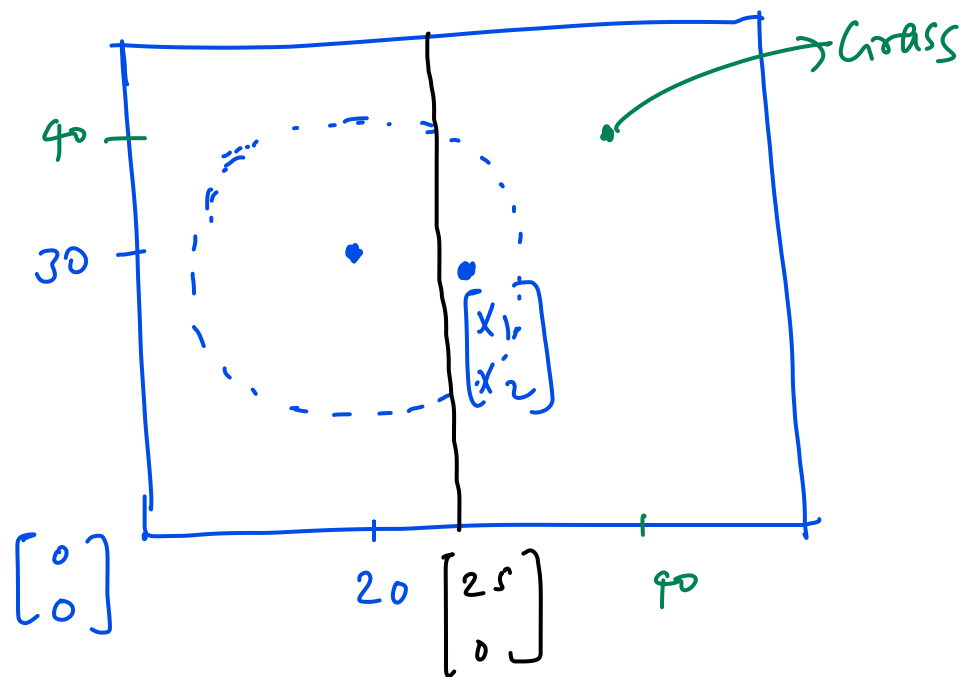
"maximum" reward.

Example:

→ A cow is at $\begin{bmatrix} 20 \\ 30 \end{bmatrix}$

→ cow is tied to a 10 unit radius rope

→ perpendicular fence that passes through $\begin{bmatrix} 25 \\ 0 \end{bmatrix}$



"How close can the cow get to the grass?"

We want to measure the distance from the grass $\begin{bmatrix} 40 \\ 40 \end{bmatrix}$.

say, cow is at $\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$

$$d\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, \begin{bmatrix} 40 \\ 40 \end{bmatrix}\right) = \sqrt{(x_1 - 40)^2 + (x_2 - 40)^2}$$

4.

$$\min_{\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}} \underbrace{(x_1 - 40)^2 + (x_2 - 40)^2}_{\text{objective function}}$$

• Rope Restriction: $(x_1 - 20)^2 + (x_2 - 30)^2 \leq 10^2 \quad \checkmark$

• Fence Restriction: $x_1 \leq 25. \quad =$

Eg(2): Suppose you're planning your lunch.

You need:

→ Enough protein for energy,

→ Limited calories so you don't overeat.

→ And you want to minimize the cost.

You can choose between two items:

→ Rice and
→ Paneer.

Let x_1 : amount of rice (in 100g units)

x_2 : amount of paneer (in 100g units)

Information (Data):

	Cost (Rs/100g)	Protein (g)	Calories
Rice	20	2	130
Paneer	60	16	260

You want:-

→ At least 40 g of protein per meal.

5. → No more than 800 kcal.

minimize total cost:

$$\min C = 20x_1 + 60x_2$$

protein restrictions: $2x_1 + 16x_2 \geq 40$

calories restrictions: $130x_1 + 260x_2 \leq 800$

$$x_1 \geq 0, x_2 \geq 0$$

General form

$\min_{x \in \mathbb{R}^d} f(x)$ — objective

constraints

$$\begin{cases} g_i(x) \leq 0 & \forall i=1, 2, \dots, k \\ h_j(x) = 0 & \forall j=1, 2, \dots, L \end{cases}$$

Inequality constraints Equality constraint

Unconstrained Optimization

6.

$$\min_{x \in \mathbb{R}} (x-5)^2$$

Solution:-

$$x=5$$

$$\hookrightarrow f'(x) = 2(x-5) = 0$$

$$f''(x) = 2 > 0$$

$$\Rightarrow \boxed{x^* = 5}$$

Pro eg:-

$$\min_{x \in \mathbb{R}} 4x^7 + 5x^5 + 3x^3 + 6x^2 + 4$$

$$f'(x) = 28x^6 + 25x^4 + 9x^2 + 12x = 0.$$

$$\min_{x \in \mathbb{R}} (x-5)^2$$

Approach:-

— Start with $x_0 \in \mathbb{R}$

for $k=1, 2, \dots, T$

— Update

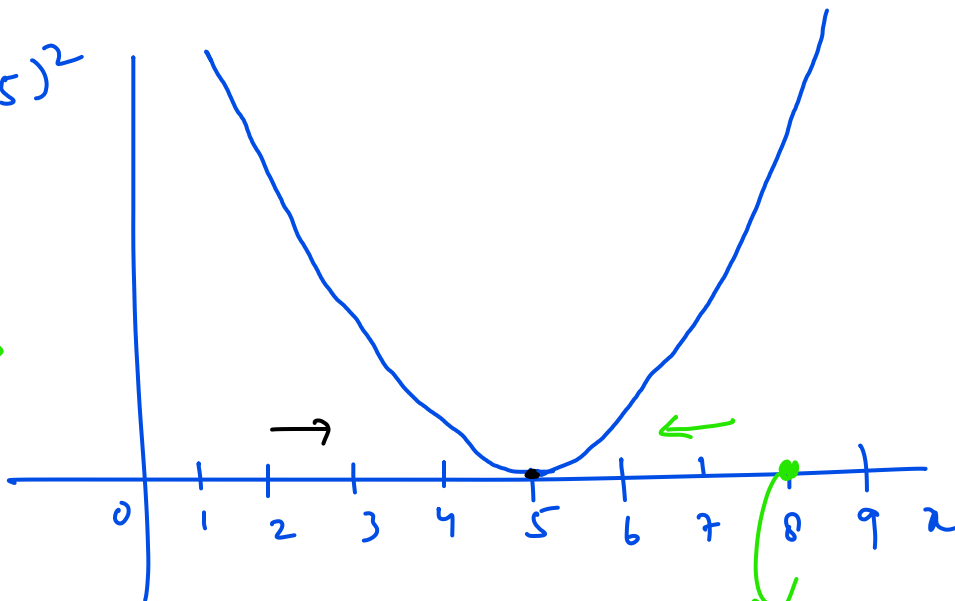
$$x_{k+1} = x_k + d_k \rightarrow \text{direction}$$

arbitrary choice

7.

or

For a given x , what is
the good direction?



$$f(x) = (x-5)^2$$

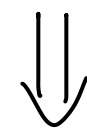
$$f'(x) = 2(x-5)$$

$$\left\{ \begin{array}{ll} \text{If } \underline{x > 5} & \text{then } \leftarrow \underline{d < 0} \\ \text{If } \underline{x < 5} & \text{then } \rightarrow \underline{d > 0} \end{array} \right\}$$

$$\hookrightarrow x > 5 \Rightarrow -f'(x) < 0$$

$$x < 5 \Rightarrow -f'(x) > 0$$

choose $d = -f'(x)$



direction will
depend on x .

i.e. d must be
a function of x .

$x_{t+1} = x_t + d$

, where $d = -f'(x)$

7

$$f(x) = (x-5)^2,$$

$$f'(x) = 2(x-5)$$

eg:- Initial guess;

8. $x_0 = 10$

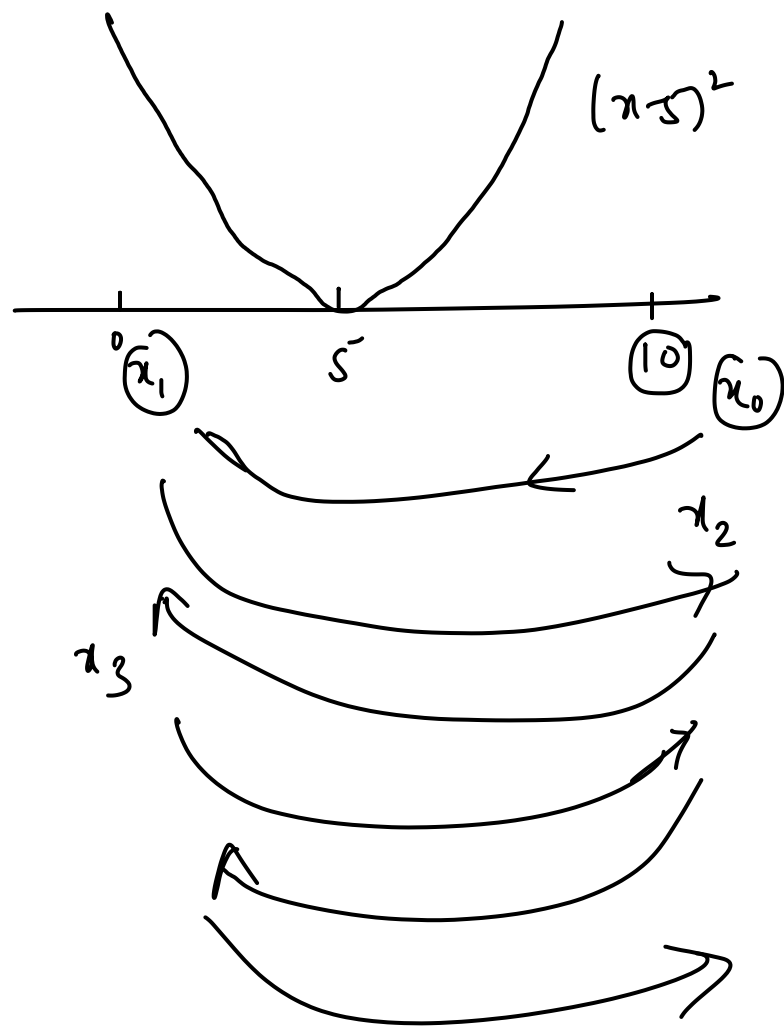
$$\begin{aligned} \underline{d} &= -f'(x_0) = -2(x_0 - 5) \\ &= -2(10 - 5) \\ &= -10 \end{aligned}$$

$$x_1 = x_0 - 10 = 10 - 10 = \underline{0}$$

$$x_2 = x_1 + d$$

$$\begin{aligned} d &= -f'(x_1) = -2(0 - 5) \\ &= 10 \end{aligned}$$

$$\underline{x_2 = 0 + 10 = 10}$$



Problem:- Not the direction but "amount".

$$x_{t+1} = x_t + d$$

$$x_{t+1} \stackrel{9.}{=} x_t - \eta_t f'(x_t)$$

→ step size (scalar quantity)
positive

How to choose step size (as a function of t)?

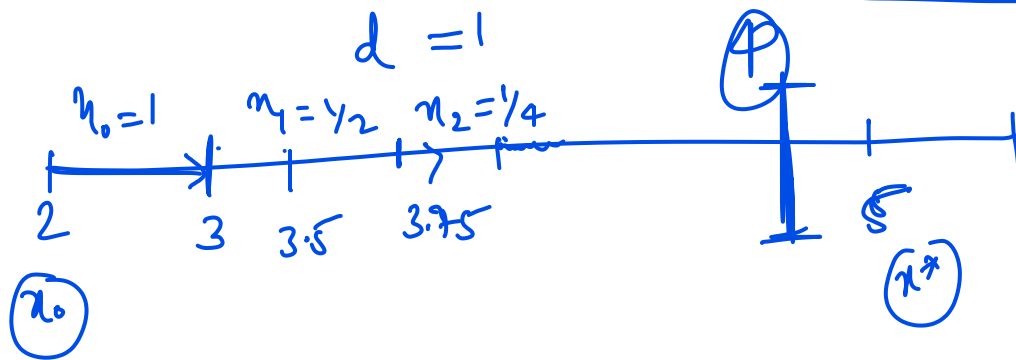
→ (D) $\eta_0 = 1, \quad \eta_1 = \frac{1}{2}, \quad \eta_2 = \frac{1}{4}, \quad \eta_3 = \frac{1}{8}, \dots$

↙ $\boxed{\eta_t = \frac{1}{2^t}}$

$$x_{t+1} = x_t + d$$

$$\boxed{x_{t+1} = x_t + \eta_t \cdot d}$$

Bad



$$3.75 + \frac{1}{8} = \dots + \frac{1}{16}$$

$$\sum_{t=0}^{\infty} \eta_t = \sum_{t=0}^{\infty} \frac{1}{2^t} = 2$$

$$\boxed{1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots = \underline{\underline{2}}}$$

take,

$$1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}, \dots$$

10.

$$\boxed{\eta_t = \frac{1}{t+1}}$$

→ good step size sequence.

$$\sum_{t=0}^{\infty} \eta_t = \sum_{t=0}^{\infty} \frac{1}{t+1} = 1 + \frac{1}{2} + \frac{1}{3} + \dots = \infty.$$

ALGORITHM - Gradient Descent Algorithm.

→ Initialize at $x_0 \in \mathbb{R} \checkmark$

for $t=1, 2, \dots$

$$x_{t+1} = x_t - \eta_t \underline{f'(x_t)}, \quad \eta_t = \frac{1}{t+1}$$

end.

Gradient-descent

11 Property:-

→ If $\eta_t = \frac{1}{t+1}$, the algorithm converges

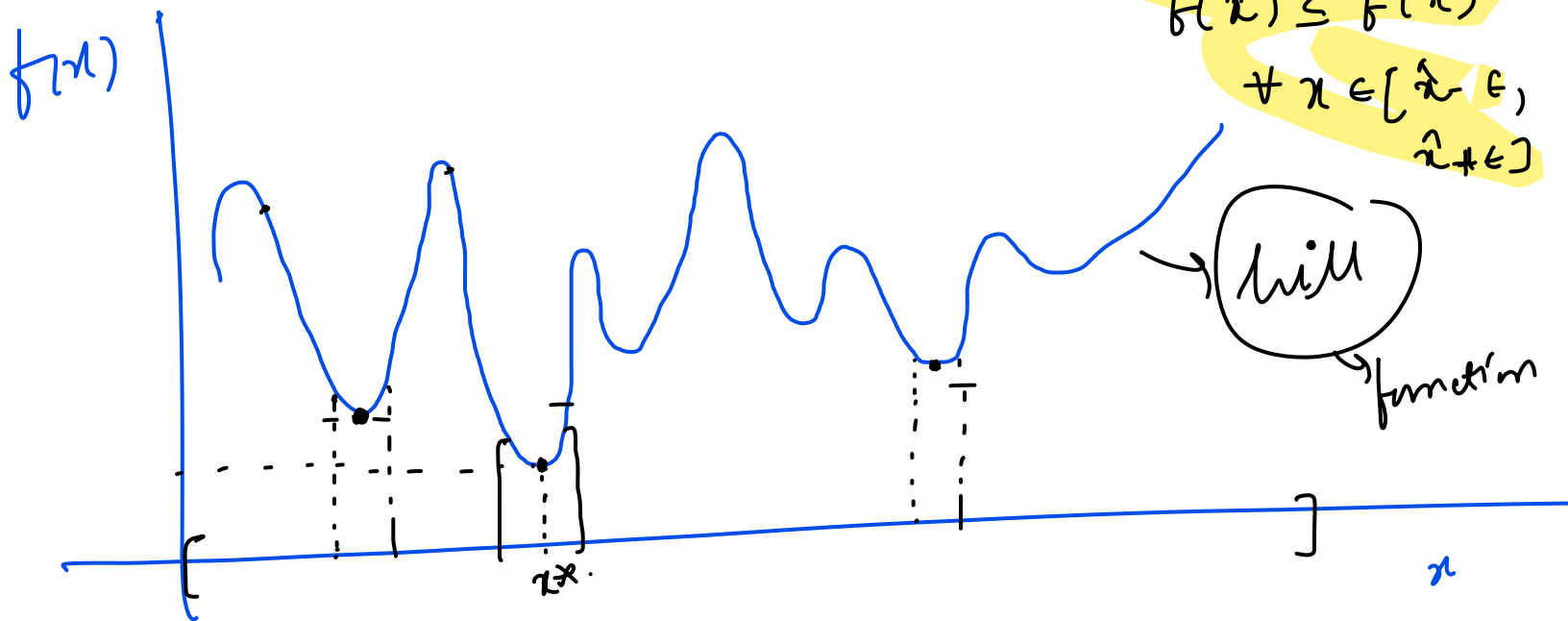
→ Gradient descent converges to "LOCAL minimum".



$\exists \epsilon > 0$ s.t.

$$f(\hat{x}) \leq f(x)$$

$$\forall x \in [\hat{x} - \epsilon, \hat{x} + \epsilon]$$



A global minimum.

$$f(x^*) \leq f(x) \quad \forall x$$

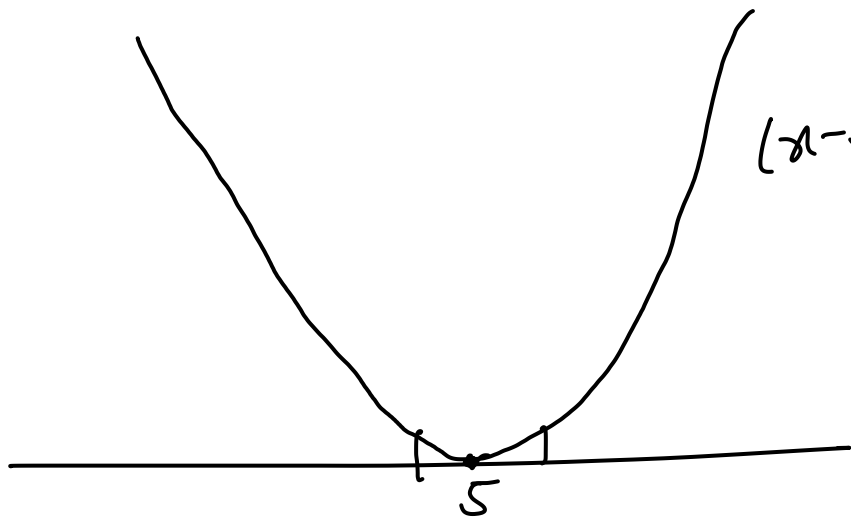
Functions in which

12.

local minimum $\Leftrightarrow$ Global minimum

convex functions

$$(x-5)^2$$



Gradient Descent

$\min_{x \in \mathbb{R}} f(x)$

Update rule:

$$x_{t+1} = x_t + \eta_t (-f'(x_t))$$

Taylor series: $f(x + \eta d) = f(x) + \eta d f'(x) + \frac{\eta^2 d^2}{2} f''(x) + \dots$

12

evaluations are all at x

$$f(x+\eta d) = f(x) + \eta d f'(x) + \underbrace{\frac{\eta^2 d^2}{2} f''(x) + \dots}_{\text{Higher order terms (X)}}$$

13.
 η small positive step size
 d direction

$$f(x+\eta d) \approx f(x) + \eta d f'(x)$$

$$\Rightarrow f(x+\eta d) - f(x) \approx \eta d f'(x)$$

Function evaluation
 at updated point
 along direction "d"

Function evaluation
 at the current
 point.

→ want 'd' such that

$$\eta d f'(x) < 0.$$

$$f(x+\eta d) - f(x) < 0$$

For the choice of

$$d = -f'(x)$$

$$d f'(x) = -(f'(x))^2 < 0$$

Higher dimensions.

$$f(x) = (x-5)^2$$

14.

$$f(x_1, x_2) = x_1^2 + 4x_2 + 8x_2^2$$

Derivative $\Leftrightarrow$ Gradient

vector of partial derivatives

$$x_{k+1} = x_k + \eta_k (-f'(x_k))$$

$$\vec{x}_{k+1} = \vec{x}_k + \eta \underbrace{(-\nabla f(x_k))}_{\text{vector}}$$

↑ ↑ ↑ ↑
vector vector scalar vector.

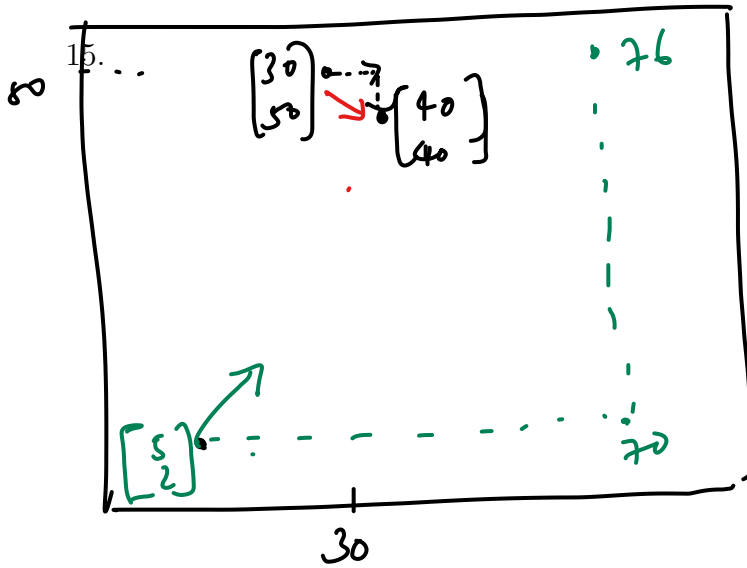
$$\nabla f\left(\begin{bmatrix} a \\ b \end{bmatrix}\right) = \begin{bmatrix} \left.\frac{\partial f}{\partial x_1}\right|_{x_1=a} \\ \left.\frac{\partial f}{\partial x_2}\right|_{x_2=b} \end{bmatrix}$$

$$f(x_1, x_2) = x_1^2 + 4x_2 + 8x_2^2$$

$\begin{bmatrix} 1 \\ 3 \end{bmatrix} \rightarrow$ initial point

$$\nabla f\left(\begin{bmatrix} 1 \\ 3 \end{bmatrix}\right) = \begin{bmatrix} 2x_1 \\ 4 + 16x_2 \end{bmatrix}_{x_1=1, x_2=3} = \begin{bmatrix} 2 \\ 52 \end{bmatrix}$$

In example of CoW -



$$d(x_1, x_2) = (x_1 - 40)^2 + (x_2 - 40)^2$$

$$\nabla d(x_1, x_2) = \begin{bmatrix} 2(x_1 - 40) \\ 2(x_2 - 40) \end{bmatrix}$$

$$\nabla d\left(\begin{bmatrix} 5 \\ 2 \end{bmatrix}\right) = \begin{bmatrix} -70 \\ -76 \end{bmatrix}$$

$$-\nabla d\left(\begin{bmatrix} 5 \\ 2 \end{bmatrix}\right) = \begin{bmatrix} 70 \\ 76 \end{bmatrix}$$

$$-\nabla d\left(\begin{bmatrix} 30 \\ 50 \end{bmatrix}\right) = \begin{bmatrix} 20 \\ -20 \end{bmatrix}$$

Eg: ①

$$\text{let } f(x) = (x-4)^2 + 7 \Rightarrow f'(x) = 2(x-4)$$

Gradient
Descent Algorithm

step size (learning rate) = 0.2

initial guess $x_0 = 1$.

→ Find x_1 and x_2 .

Soln:-

$$x_{t+1} = x_t + \eta (-f'(x_t))$$

$$x_{t+1} = x_t - \eta f'(x_t)$$

$$x_1 = x_0 - \eta \underbrace{f'(x_0)}$$

$$= 1 - 0.2 \times (-6)$$

$$\boxed{x_1 = 2.2}$$

$$x_2 = x_1 - \eta f'(x_1)$$

$$= 2.2 - 0.2(-3.6)$$

$$\boxed{x_2 = 2.92}$$

$$f'(x) = 2(x-4)$$

$$f'(x_0) = 2(1-4)$$

$$= -6$$

$$f'(x_1) = 2(x_1-4)$$

$$= 2(2.2-4)$$

$$= -3.6$$

→ what is the value of objective function after second

Iteration?

$$f(x_2) = (2.92-4)^2 + 7 =$$

eg (2):-

$$f(x) = x^3 - 3x^2 + 2$$

17.

$$\text{step size (learning rate)} = 0.05$$

$$\text{initial guess } x_0 = \underline{0.5}$$

→ what is the value of x_1 and x_2 ?

soln:-

$$x_{t+1} = x_t - \eta f'(x_t)$$

$$f'(x) = 3x^2 - 6x$$

$$x_1 = \underline{x_0} - \underline{0.05} f'(x_0) = 0.5 - 0.05 (3(0.5)^2 - 6(0.5))$$

$$= \underline{0.5 - 0.05(-2.25)} = \underline{0.6125}$$

$$x_2 = x_1 - 0.05 f'(x_1)$$

$$= 0.6125 - 0.05 [3(0.6125)^2 - 6(0.6125)]$$

$$x_2 = \underline{0.7399}$$

eg (3): - $f(x, y) = x^2 + 4y^2$ ✓

18.

stepsize (learning rate) = 0.1

initial guess $(x_0, y_0) = (1, 1)$

Soln:- $\vec{x}_{t+1} = \vec{x}_t - \eta \nabla f'(\vec{x}_t)$

$$\nabla f(x, y) = \begin{bmatrix} 2x \\ 8y \end{bmatrix} \quad = \begin{bmatrix} 2 \\ 8 \end{bmatrix}$$

$x=1, y=1$

$$\begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = \begin{bmatrix} x_0 \\ y_0 \end{bmatrix} - 0.1 \begin{bmatrix} 2 \\ 8 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ 1 \end{bmatrix} - \begin{bmatrix} 0.2 \\ 0.8 \end{bmatrix} = \begin{bmatrix} 0.8 \\ 0.2 \end{bmatrix} \quad \checkmark$$

Now,

$$\nabla f(x, y) = \begin{bmatrix} 2x \\ 8y \end{bmatrix}$$

$$\nabla f(x_1, y_1) = \begin{bmatrix} 2x \\ 8y \end{bmatrix} \quad = \begin{bmatrix} 1.6 \\ 1.6 \end{bmatrix}$$

$x_1 = 0.8, y_1 = 0.2$ ✓

$$\begin{bmatrix} x_2 \\ y_2 \end{bmatrix} = \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} - 0.1 \begin{bmatrix} 1.6 \\ 1.6 \end{bmatrix} \quad \checkmark$$

$$= \begin{bmatrix} 0.8 \\ 0.2 \end{bmatrix} - \begin{bmatrix} 0.16 \\ 0.16 \end{bmatrix} = \begin{bmatrix} 0.64 \\ 0.04 \end{bmatrix} \quad \checkmark$$

$$f(x + \eta d) = f(x) + \eta d f'(x) + \frac{\eta^2 d^2}{2} f''(x) + \dots$$

Higher order Taylor series:

$$f(x + \eta d) = f(x) + \eta d^T \nabla f(x) + \dots$$

↑
vector

↑
vector

$$f(x + \eta d) - f(x) \approx \eta d^T \nabla f(x)$$

want d s.t

$$d^T \nabla f(x) < 0$$

Note! ∇f $d = -\nabla f(x)$

20.

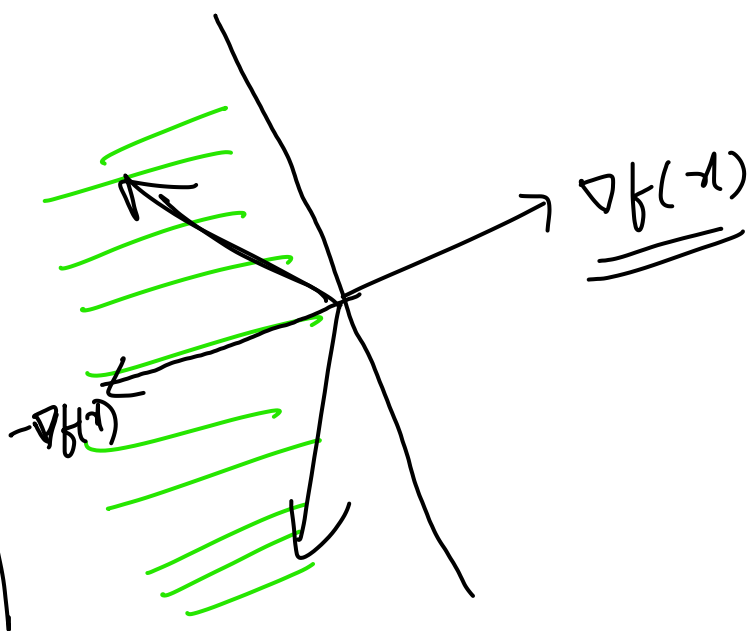
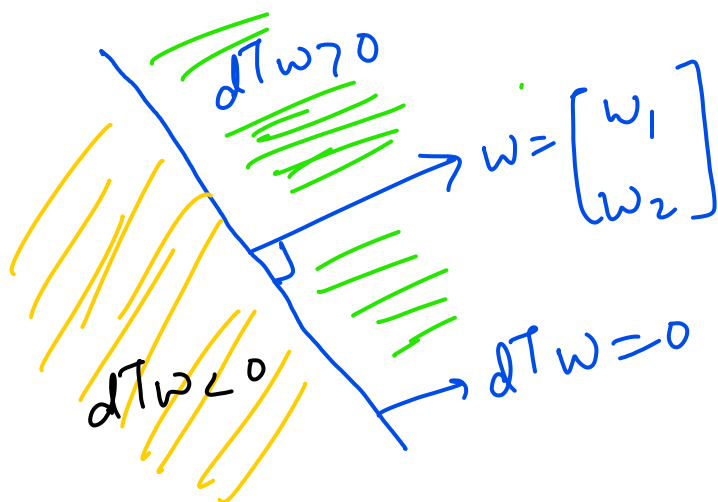
$$a^T a = \|a\|^2$$

$$(-\nabla f(x))^T (\nabla f(x))$$

$$= -\|\nabla f(x)\|_2^2 < 0$$

suppose

$$d^T w = 0?$$

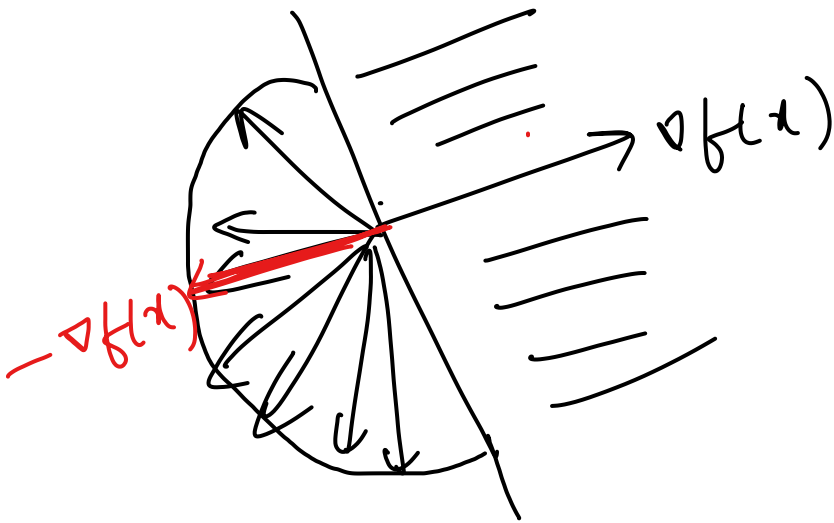


$-\nabla f(x)$ gives the "steepest descent".

$$d = -\nabla f(x)$$

Gradient Descent is also called "steepest descent".

21.



$$\min_{x \in \mathbb{R}^d} f(x)$$

$$g(x) \leq 0$$

$$(\underline{x_1 - 20})^2 + (\underline{x_2 - 30})^2 \leq 10^2$$

22.

23.

24.

25.

26.

27.

28.

29.

30.